

COST MINIMIZATION PROBLEM SET

ECON101—INTRODUCTION TO MICROECONOMICS.

FALL 2024

PROF. ALEJANDRO MELO PONCE

NAZARBAYEV UNIVERSITY

Problem 1

Poizy the gardener is an avid gardening enthusiast. He has found that the number of happy plants, h , depends on the amount of light l , and water, w . In fact, Poizy noticed that plants require twice as much light as water, and any more or less is wasted. Thus, Poizy's production function is $h = \min\{l, 2w\}$.

- (a) Suppose that Poizy is using 1 unit of light. What is the least amount of water he can use and still produce a happy plant?
- (b) Suppose that Poizy wants to produce 4 happy plants. What are the minimum amounts of light and water required?
- (c) What is Poizy's conditional input demand function for light? What is Poizy's conditional input demand function for water?
- (d) If each unit of light costs w_1 and each unit of water costs w_2 , Poizy's cost function $C(w_1, w_2, h)$ is?

Problem 2

Poizy, from the previous task, is also a NU student who is currently taking a WCS course. It turns out that his instructor, Chatterbox Chucklehead (CC), is also an avid gardener. However, CC uses an alternative method of gardening. CC has found that happy plants only need fertilizer and *talk*. In an amazing discovery, CC has figured out that talk and fertilizer are perfect substitutes! Thus, suppose that f is the number of bags of fertilizer used and t is the number of hours CC talks to the plants; then the number of happy plants produced is exactly $h = t + 2f$. Suppose fertilizer costs w_f per bag and talk costs w_t per hour.

- (a) If CC uses no fertilizer, how many hours of talk must be devoted if CC wants one happy plant?
- (b) If CC doesn't talk to the plants at all, how many bags of fertilizer will be needed for one happy plant?
- (c) If $w_t < w_f/2$, would it be cheaper for CC to use fertilizer or talk to raise one happy plant?
- (d) What is CC's cost function $c(w_f, w_t, h)$?
- (e) What is CC's conditional input demand for talk, $t(w_f, w_t, h)$?

Problem 3

NU's cafeteria produces meals, using two inputs, "gas burners" g , and a mystery ingredient x . I am not at liberty to say what the mystery ingredient is, but I have on good authority that "meat and fungus are involved." The cafeteria's production function is $f(x, g) = gx^2$, where x is the amount of the mystery ingredient, g are gas burners, and $f(x, g)$ is the amount of meals produced.

- (a) Does the cafeteria have increasing, constant, or decreasing returns to scale.
- (b) For the rest of the problem, suppose that we are considering the short-run, so that the amount of gas burners is fixed at $g_0 = 10$. How many units of the mystery ingredient x takes to produce 1440 meals? If the input costs are w_x per unit of x and w_g for running each burner, what is the short-run cost to produce 1440 meals?
- (c) How many units of x are needed to produce y meals? If input costs are w_x and w_g , what is the cost to produce y means?
- (d) Provide expressions for the average total cost, average variable cost, average fixed cost, and marginal cost curves as functions of y , w_x and w_g .